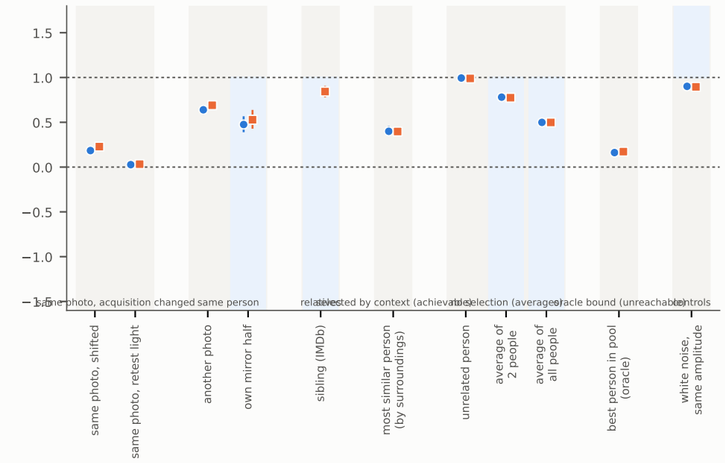
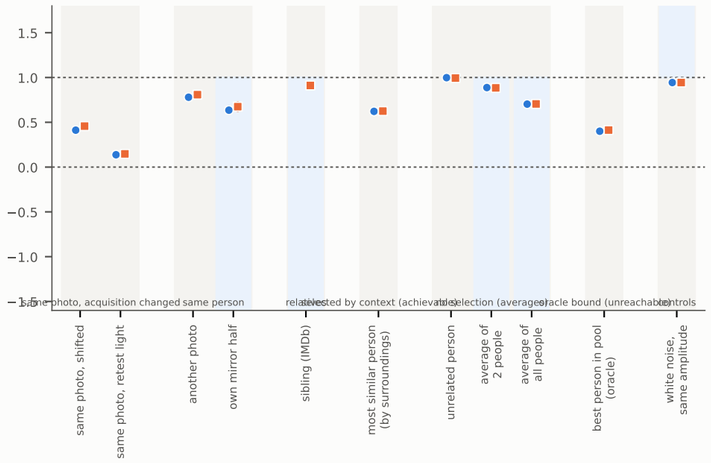
faces, 16 px windows
CelebA
IMDb-WIKI

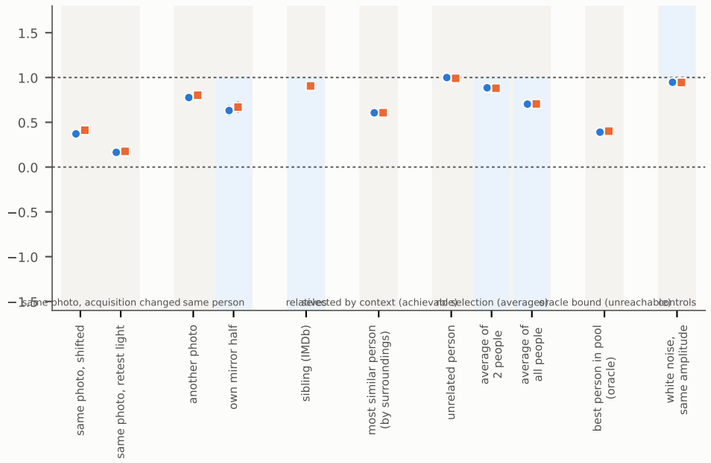
MSE: expectations met CelebA 5/6, IMDb-WIKI 6/7



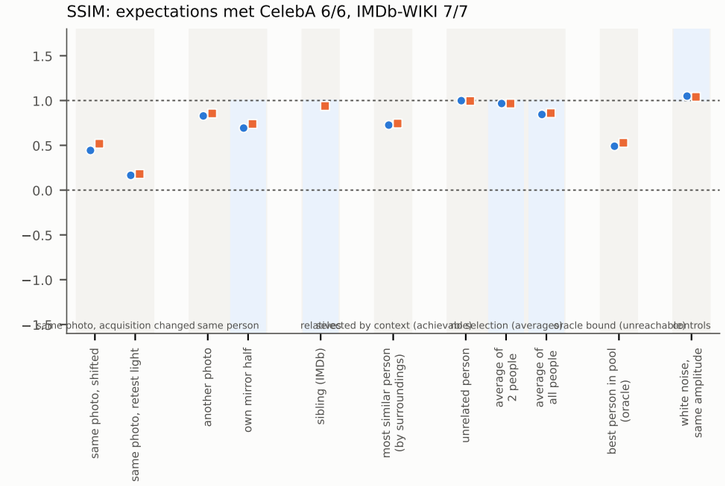
RMSE: expectations met CelebA 5/6, IMDb-WIKI 6/7



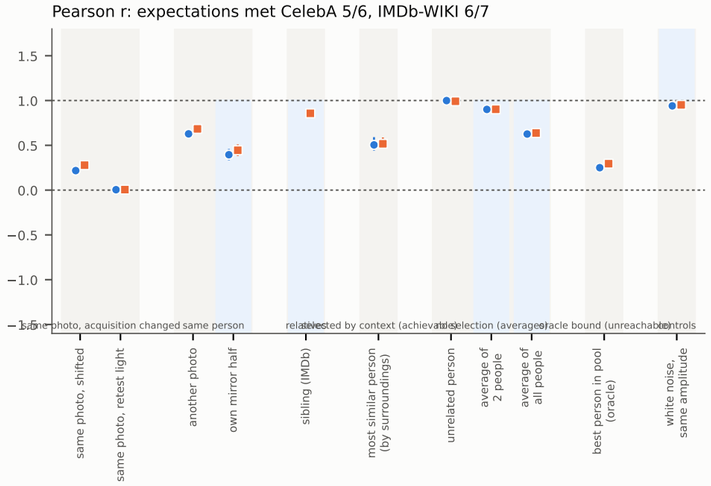
MAE: expectations met CelebA 5/6, IMDb-WIKI 6/7



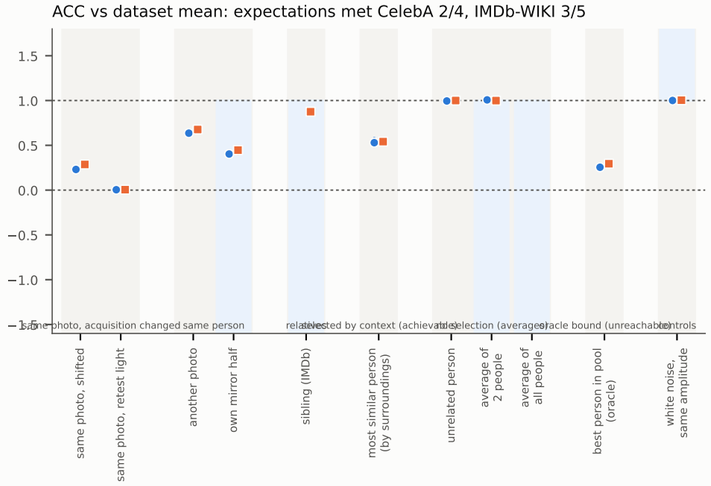
SSIM: expectations met CelebA 6/6, IMDb-WIKI 7/7



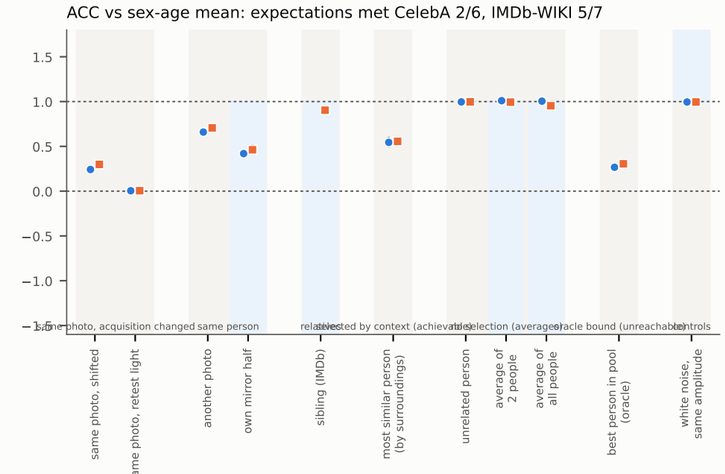
Pearson r: expectations met CelebA 5/6, IMDb-WIKI 6/7



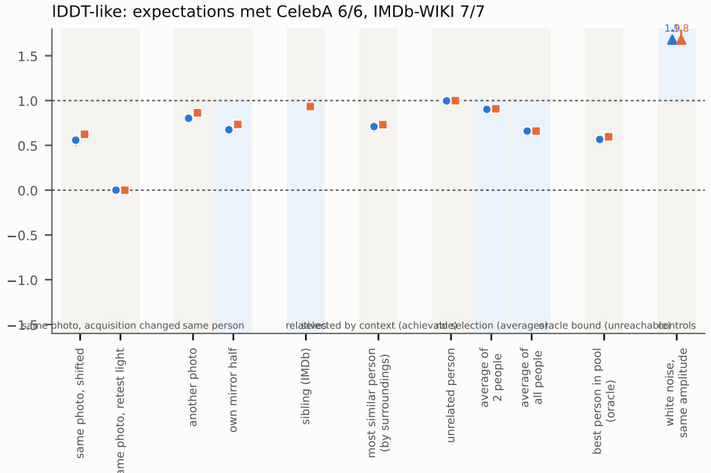
ACC vs dataset mean: expectations met CelebA 2/4, IMDb-WIKI 3/5



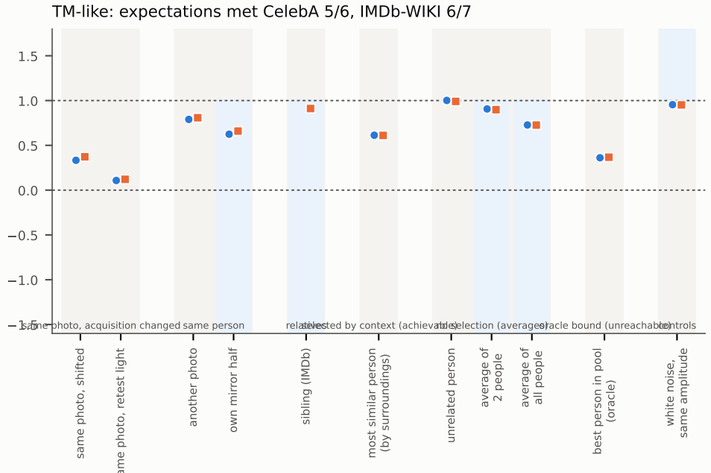
ACC vs sex-age mean: expectations met CelebA 2/6, IMDb-WIKI 5/7



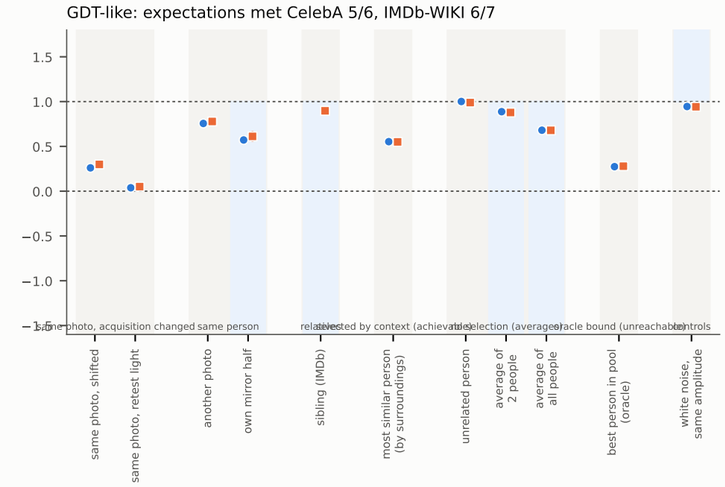
IDDT-like: expectations met CelebA 6/6, IMDb-WIKI 7/7



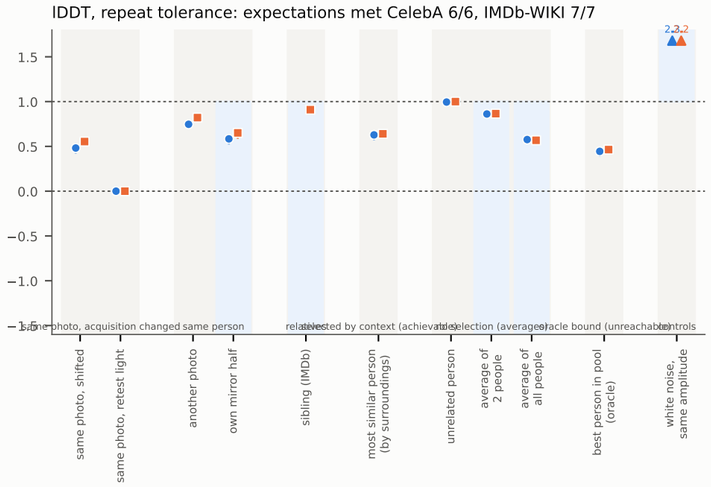
TM-like: expectations met CelebA 5/6, IMDb-WIKI 6/7



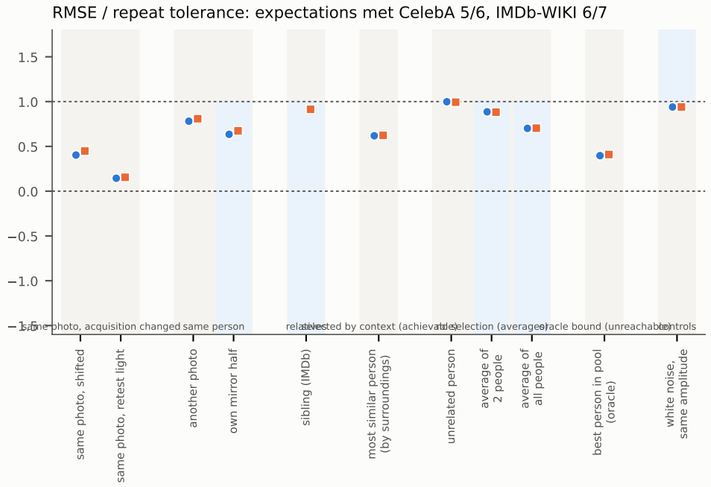
GDT-like: expectations met CelebA 5/6, IMDb-WIKI 6/7



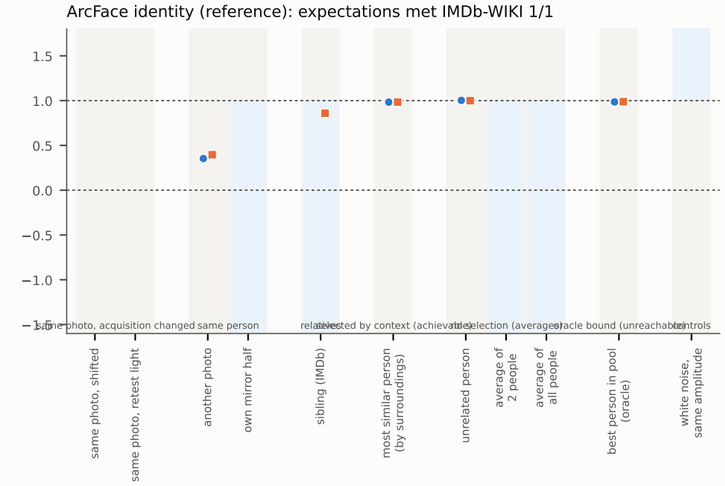
IDDT, repeat tolerance: expectations met CelebA 6/6, IMDb-WIKI 7/7



RMSE / repeat tolerance: expectations met CelebA 5/6, IMDb-WIKI 6/7



ArcFace identity (reference): expectations met IMDb-WIKI 1/1



Reality-B expectations (value = R)

	CelebA: E1 mirror<str	CelebA: E2 sib<str	CelebA: E3B mean<str	CelebA: E3B mean2<str	CelebA: E3B order by k	CelebA: E5B wrong<str	CelebA: E6B noise>str	IMDb-WIKI: E1 mirror<str	IMDb-WIKI: E2 sib<str	IMDb-WIKI: E3B mean<str	IMDb-WIKI: E3B mean2<str	IMDb-WIKI: E3B order by k	IMDb-WIKI: E5B wrong<str	IMDb-WIKI: E6B noise>str
MSE	pass 0.48	n/a	pass 0.50	pass 0.78	pass 0.5/0.8/1.0	pass 0.98	fail 0.90	pass 0.53	pass 0.84	pass 0.50	pass 0.77	pass 0.5/0.8/1.0	pass 0.98	fail 0.90
RMSE	pass 0.64	n/a	pass 0.70	pass 0.89	pass 0.7/0.9/1.0	pass 0.98	fail 0.94	pass 0.67	pass 0.91	pass 0.70	pass 0.88	pass 0.7/0.9/1.0	pass 0.99	fail 0.94
MAE	pass 0.63	n/a	pass 0.70	pass 0.89	pass 0.7/0.9/1.0	pass 0.98	fail 0.95	pass 0.67	pass 0.91	pass 0.71	pass 0.88	pass 0.7/0.9/1.0	pass 0.98	fail 0.95
PSNR	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
PSNR, fixed range	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
SSIM	pass 0.69	n/a	pass 0.84	pass 0.97	pass 0.8/1.0/1.0	pass 1.01	pass 1.05	pass 0.74	pass 0.94	pass 0.86	pass 0.97	pass 0.9/1.0/1.0	pass 1.01	pass 1.04
Pearson r	pass 0.40	n/a	pass 0.63	pass 0.90	pass 0.6/0.9/1.0	pass 0.99	fail 0.94	pass 0.44	pass 0.86	pass 0.64	pass 0.90	pass 0.6/0.9/1.0	pass 1.00	fail 0.95
ACC vs dataset mean	pass 0.40	n/a	n/a	fail 1.01	n/a	pass 1.00	fail 1.00	pass 0.45	pass 0.88	n/a	fail 1.00	n/a	pass 1.01	fail 1.00
ACC vs sex-age mean	pass 0.42	n/a	fail 1.00	fail 1.01	fail 1.0/1.0/1.0	pass 1.00	fail 1.00	pass 0.46	pass 0.90	pass 0.95	fail 0.99	pass 1.0/1.0/1.0	pass 1.01	fail 1.00
IDDT-like	pass 0.67	n/a	pass 0.66	pass 0.90	pass 0.7/0.9/1.0	pass 1.04	pass 1.91	pass 0.73	pass 0.93	pass 0.66	pass 0.91	pass 0.7/0.9/1.0	pass 1.05	pass 1.80
TM-like	pass 0.62	n/a	pass 0.73	pass 0.91	pass 0.7/0.9/1.0	pass 0.97	fail 0.95	pass 0.66	pass 0.91	pass 0.73	pass 0.90	pass 0.7/0.9/1.0	pass 0.97	fail 0.95
GDT-like	pass 0.57	n/a	pass 0.68	pass 0.89	pass 0.7/0.9/1.0	pass 0.97	fail 0.94	pass 0.61	pass 0.90	pass 0.68	pass 0.88	pass 0.7/0.9/1.0	pass 0.97	fail 0.94
IDDT, repeat tolerance	pass 0.58	n/a	pass 0.58	pass 0.86	pass 0.6/0.9/1.0	pass 1.06	pass 2.33	pass 0.65	pass 0.91	pass 0.57	pass 0.87	pass 0.6/0.9/1.0	pass 1.08	pass 2.21
RMSE / repeat tolerance	pass 0.63	n/a	pass 0.70	pass 0.89	pass 0.7/0.9/1.0	pass 0.98	fail 0.94	pass 0.67	pass 0.91	pass 0.70	pass 0.88	pass 0.7/0.9/1.0	pass 0.99	fail 0.94

R = (d - d_self) / (d_stranger - d_self): 0 = the target itself, 1 = unrelated person (mean of 16; Amendment F1). Bars: 95% interval, bootstrap over families and window positions. Blue bands: reality-B expectations (mirror, sibling, averages below 1; wrong detail at or above 0.9; noise above 1).
face frame _v6f: visual QC approved by Kate, 6 Nov 2026 (Amendment 10); landmark affine to the population template (face protocol, Limitations 3)